Large Reasoning Models

Reasoning, Prompting Techniques, and Instruction Tuning

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- **Why study LRMs?**
  - LLMs like GPT-3 and GPT-4 excel at generating text.
  - However, they often rely on **pattern matching**, not true reasoning.
  - Real-world applications like **math problem-solving**, **code synthesis**, and **scientific hypothesis generation** require multi-step reasoning.
- **Examples:**
  - *The same model can answer a multi-step problem far better when asked to work through it: appending "Let's think step by step" raises InstructGPT from 17.7% to 78.7% on MultiArith. The knowledge was already in the weights; only the prompt changed.*
  - *DeepMind's AlphaCode reached competition-level programming not by reasoning its way to one answer, but by sampling a very large number of candidate programs and filtering them against the example tests, averaging the top 54.3% of human entrants on Codeforces.*

Learning Outcomes

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By the end of this session, you should be able to:

- Define and distinguish Large Reasoning Models (LRMs) from simpler pattern-matching LLMs.
- Name the kinds of reasoning an LRM is asked to perform, and recognise which one a given task calls for.
- Understand advanced reasoning techniques, including Chain-of-Thought (CoT), Tree-of-Thought (ToT), and self-consistency methods.
- Explain how instruction tuning and reinforcement learning (RL) can refine and enhance reasoning abilities in language models.
- Critically assess the limits of current reasoning models, including whether a reasoning trace can be read as an account of how the answer was reached.

What Are Large Reasoning Models (LRMs)?

- **Definition:**
  - Large Language Models (LLMs) optimized for **multi-step logical reasoning**
  - Emphasize Chain-of-Thought (CoT) style reasoning
- **Key Examples:**
  - OpenAI: o1, o3
  - Google: Gemini 2.0 Flash Thinking
  - DeepSeek: R1
  - QwQ (Alibaba)
  - Claude 3.7 Sonnet (extended thinking)
  - Magistral (Mistral)

- **Distinctive Behavior:**
  - "Thinking" before answering rather than emitting the answer directly
  - Intermediate reasoning traces, either shown to the user or kept hidden
  - Step-by-step intermediate computations
  - More decode steps spent on a hard question than an easy one
  - Better performance on symbolic tasks and math
  - Often paired with advanced prompting or finetuning

What Are Large Reasoning Models (LRMs)? (cont.)

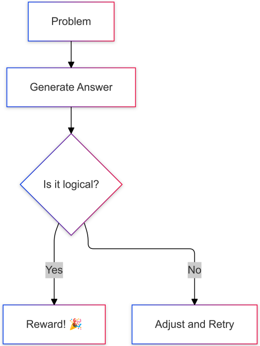
- **Examples:**
  - **PaLM + CoT Prompting:**
    Chain-of-Thought prompting raised PaLM 540B on GSM8K (grade-school math) from 17.9% to 56.9%, without changing a single weight.
  - **Gemini 2.0 Flash Thinking:**
    Exposes the reasoning steps behind an answer instead of hiding them.
  - **DeepSeek R1:**
    Trained with large-scale reinforcement learning on problems whose answers can be checked automatically.
    Its predecessor R1-Zero developed an "aha moment" during training: it began pausing mid-solution to re-examine its own approach, without ever being shown an example of doing so.

Large language models (LLMs) and large reasoning models (LRMs) share foundational technologies but serve different purposes:

- **Core Function:**
  - **LLMs:** Learn patterns in data to generate fluent, human-like text.
  - **LRMs:** Extend LLMs to solve problems requiring logical reasoning and contextual understanding.
- **Use Cases:**
  - **LLMs:** Content generation, language translation, summarization, and answering user queries.
  - **LRMs:** Math problem solving, interpreting ambiguous clinical data, and complex decision-making.

LLMs vs. LRMs (cont.)

- **Reasoning Ability:**
  - **LLMs:** Limited structured reasoning; may struggle with multi-step logic or ambiguity.
  - **LRMs:** Trained to apply consistent reasoning steps; outputs are more logical and verifiable.
- **Performance:**
  - **LLMs:** Fast response times; optimized for scalability and speed.
  - **LRMs:** Slower response times; require more processing to reason through problems step by step.
- **Best Suited For:**
  - **LLMs:** High-volume, low-risk tasks where speed is critical.
  - **LRMs:** High-stakes or complex tasks where accuracy, explainability, and logic matter.



The reward loop behind the RL stage below: a candidate answer is checked, rewarded when the reasoning holds up, and revised when it does not. Source: Chaddha (2025), "Large Reasoning Models: The AI That Solves Problems Like a Human (But 10x Faster)," Medium; the author's own illustrative diagram, not from a paper.

How Do LRMs Work? (cont.)

LRMs use a combination of training methods and prompt strategies to enhance the reasoning capabilities of LLMs.

- **Training on Enriched Datasets**
  - Includes reasoning-focused examples
  - Real-world scenarios with clear outcomes
  - Teaches both correct answers and reasoning steps
- **Reinforcement Learning (RL)**
  - Rewards for logical, correct answers
  - Penalties for incorrect or inconsistent reasoning
  - Reinforces desirable reasoning patterns
- **Human Feedback (RLHF)**
  - Human reviewers guide model outputs

How Do LRMs Work? (cont.)

- Refines nuanced reasoning strategies
- Useful in domains needing expertise (e.g., healthcare, finance)

- **Prompt Engineering**
  - Carefully designed prompts activate reasoning
  - Guides model through multi-step or layered tasks
  - Generic prompts may not trigger deep reasoning
- **Chain-of-Thought (CoT) Prompting**
  - Breaks problems into smaller steps
  - Explores multiple reasoning paths
  - Improves transparency and accuracy
  - Essential for auditability and explainability

Types of Reasoning in LRMs

Large reasoning models (LRMs) employ several types of reasoning to simulate or support human-like problem solving:

- **Deductive Reasoning**
  - Applies general rules to specific cases for logically certain conclusions (top-down reasoning).
  - Excels at structured, rule-based tasks requiring high logical consistency.
  - Example domains: regulatory compliance, medical protocol analysis.
  - *Example:*
    Rule: All patients with a fever and cough should be tested for influenza.
    Case: Patient A has a fever and cough.
    Deduction: Patient A should be tested for influenza.

Types of Reasoning in LRMs (cont.)

- **Inductive Reasoning**
  - Draws general conclusions from specific observations in data.
  - Supports probabilistic predictions and generalization to unseen inputs.
  - Useful in dynamic, data-rich scenarios (e.g., fraud detection).
  - *Example:*
    Observation: 90% of emails with suspicious links are phishing attempts.
    Induction: An email with a suspicious link is likely to be a phishing attempt.

- **Abductive Reasoning**
  - Infers the most likely explanation from incomplete or uncertain evidence.
  - Generates hypotheses rather than definitive answers.
  - Effective in domains like medical diagnostics, where plausible interpretations are needed.
  - *Example:*
    Evidence: The lawn is wet in the morning.
    Possible explanations: It rained overnight, or the sprinkler was on.
    Abduction: If the weather report says no rain, the most likely explanation is the sprinkler was on.

Types of Reasoning in LRMs (cont.)

- **Analogical Reasoning**
  - Identifies similarities between different situations or datasets.
  - Transfers insights from one context to another for novel problem solving.
  - Example: recommending products based on similar customer behavior.
  - *Example:*
    Situation A: A student learns to solve quadratic equations by factoring.
    Situation B: The student applies the same factoring method to solve cubic equations.
    Analogy: The approach used for quadratics is adapted to cubics.

Reasoning vs. Pattern-Matching in LLMs

Reasoning vs. Pattern-Matching in LLMs

- **Pattern-Matching Models**
  - **How it works:** LLMs are trained on vast amounts of text data to learn statistical associations between words and phrases.
  - **Characteristics:**
    - Generate text based on learned patterns without deeper understanding.
    - Often produce fluent and coherent text.
    - May struggle with complex reasoning tasks or multi-step logic.
  - **Example:** Given the prompt "The capital of France is", the model outputs "Paris" by recalling frequent patterns in the training data.
  - **Limitations:**
    - May produce plausible-sounding but incorrect answers if the pattern is misleading.
    - Struggles with tasks that require connecting multiple facts or reasoning through steps.

Reasoning vs. Pattern-Matching in LLMs (cont.)

- **Reasoning-Focused Models**
  - **How it works:** LRMs extend LLMs by incorporating structured reasoning techniques.
  - **Characteristics:**
    - Generate structured intermediate steps to arrive at conclusions.
    - Use multi-hop inference to connect information across several statements or facts.
    - Can break down complex problems into smaller, logical steps.
  - **Example:** To answer "If Alice has 3 apples and buys 2 more, how many does she have?", the model first adds 3 and 2, then states the answer.
  - **Benefits:**
    - Provides explanations or shows their thought process, making answers more transparent.
    - Better at tasks that require logic, calculation, or combining information from different sources.

Reasoning vs. Pattern-Matching in LLMs (cont.)

- **Examples**
  - “Which is heavier: a kilogram of feathers or a kilogram of iron?”

    A pattern-matching model may be misled by the association of “iron” with heaviness.

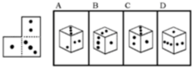
    A reasoning model will recognize that both weigh the same.

  - In logic puzzles, pattern-based models tend to contradict earlier premises

    Reasoning-aware models (e.g., using Chain-of-Thought prompting) remain consistent and logical.

Reasoning vs. Pattern-Matching in LLMs (cont.)

Still Terrible



Box folding: F
LLMs cannot solve box-folding problems and struggle with abstract figures and 3d spaces.

Abstract Map Navigation: D
LLMs cannot solve complex problems like route-planning in abstract 2d spaces.


Venn diagram fail

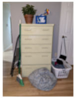

Supposed to be an unfolded box

Drawing diagrams: F
All tested models are comically bad at producing diagrams to exact, or even loose specifications.

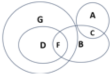
Improving

Stacking objects: B+
Claude solved a puzzle GPT failed on a year ago. Progress!

Real Map Navigation: C+
Models showed uneven but improving performance in multi-stop navigation.



Spatial reasoning in photos: B
Models seem to do better reasoning from photos of real scenes than from abstractions.



Venn Diagrams: B
Claude and GPT can reason from verbal descriptions of Venn diagrams.

Spatial reasoning is a case where the split is visible task by task: models remain poor on abstract figures and on drawing diagrams, and are improving on scenes and real maps. Source: Bos (2024), Towards Data Science: the author's own informal testing of GPT-4 and Claude 3.5 Sonnet, July 2024, not a benchmark.

- **Empirical Separation: When Does Reasoning Matter?**
  - On simple factual questions, both model types may perform well.
  - For challenging tasks like GSM8K (grade-school math), symbolic reasoning, or logic puzzles, pattern-matching alone is not enough.
  - Success on these benchmarks improves dramatically when models are prompted to reason step by step (e.g., using Chain-of-Thought prompting).
  - *Example:* In multi-step math problems, models that show their work (reasoning-focused) achieve much higher accuracy than those that just guess the answer.
  - This highlights the need for explicit reasoning capabilities in advanced language models.

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Following are some of the leading Large Reasoning Models (LRMs) as of 2024/2025:

- **OpenAI o1/o3**
  - **Architecture:** Undisclosed; trained with reinforcement learning to reason before answering.
  - **Training Data Transparency:** Proprietary, limited details.
  - **Strengths:** High-quality reasoning, STEM, coding.
  - **Reasoning Approach:** Hidden chain of thought, with the effort budget selectable as low, medium, or high.
  - **Efficiency:** 200K context window, 100K maximum output tokens.
  - **Best Use Case:** General AI, STEM research, API function calling.

DeepSeek R1

- **Architecture:** Mixture of Experts (MoE, 671B total, 37B active).
- **Training Data Transparency:** Open weights (MIT licence); base model pretrained on 14.8T tokens and the RL recipe described in the paper.
- **Strengths:** Math, coding, and STEM reasoning at open-weight cost.
- **Reasoning Approach:** Large-scale RL on problems whose answers can be checked automatically, producing long explicit chains of thought.
- **Efficiency:** Only 37B of 671B parameters are active per token, so each token costs a fraction of a dense model of the same size.
- **Best Use Case:** Self-hosted reasoning assistant, coding and debugging.

Leading Large Reasoning Models (cont.)

- **Gemini 2.0 (Flash Thinking)**
  - **Architecture:** Multimodal Transformer (Text+Image+Speech).
  - **Training Data Transparency:** Proprietary, multimodal (text, images, APIs).
  - **Strengths:** Multimodal (images, text, speech), deep reasoning.
  - **Reasoning Approach:** Flash Thinking mode (transparent thought process).
  - **Efficiency:** 1M token context (experimental), optimized for speed.
  - **Best Use Case:** AI assistant (multimodal, research, planning).

Claude 3.7 Sonnet

- **Architecture:** Undisclosed; a single model that answers either immediately or after extended thinking.
- **Training Data Transparency:** Proprietary; evaluations documented in a published system card.
- **Strengths:** Coding, agentic tasks, long-form dialogue.
- **Reasoning Approach:** Extended thinking, shown to the user rather than hidden.
- **Efficiency:** 200K context; thinking can be switched off, and its token budget set per API call.
- **Best Use Case:** One deployment covering both quick replies and hard problems.

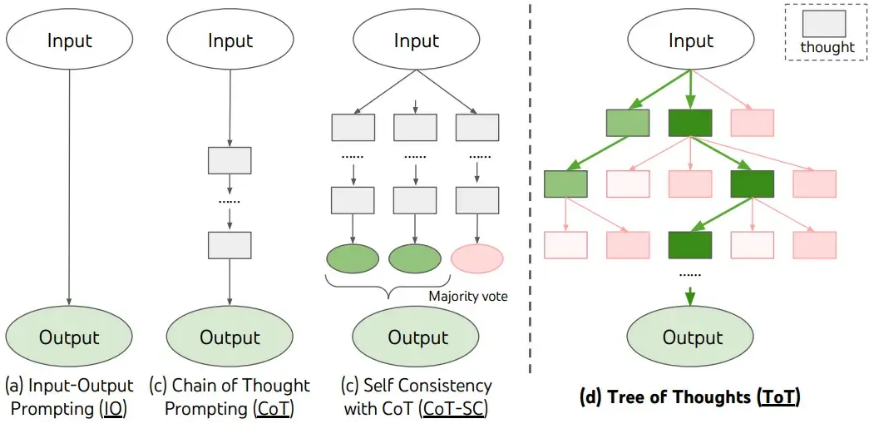
Magistral (Mistral)

- **Architecture:** Dense Transformer; Magistral Small at 24B, Magistral Medium for enterprise use.
- **Training Data Transparency:** Magistral Small released open-weight under Apache 2.0, with the RL recipe published.
- **Strengths:** Reasoning traces that stay in the user's own language across scripts and alphabets.
- **Reasoning Approach:** RL-trained chains of thought, kept legible for review.
- **Efficiency:** 24B open-weight variant, self-hostable.
- **Best Use Case:** Regulated or multilingual settings where the trace must be auditable.

Leading Large Reasoning Models (cont.)

- **QwQ (Alibaba)**
  - **Architecture:** Dense Transformer (32B), RL-trained for reasoning.
  - **Training Data Transparency:** Open weights under Apache 2.0; training corpus undisclosed.
  - **Strengths:** Math, logical puzzles, chain-of-thought self-verification.
  - **Reasoning Approach:** Outcome-reward RL scaled up from a cold-start checkpoint, plus self-reflection.
  - **Efficiency:** 32B parameters, yet comparable to DeepSeek-R1 on several reasoning benchmarks.
  - **Best Use Case:** Mathematical proofs, advanced reasoning on modest hardware.

Reasoning Techniques in LLMs



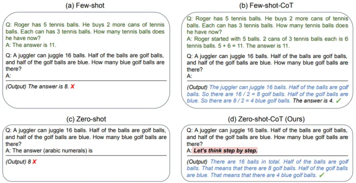
Input-output prompting, Chain-of-Thought, Self-Consistency and Tree-of-Thought side by side. Source: Yao et al. (2023), Fig. 1.

Chain-of-Thought Prompting (Recap)

- CoT prompting makes the model generate intermediate reasoning steps before the final answer, either by showing worked examples or by appending "Let's think step by step".
- For reasoning models this is the central mechanism: the techniques that follow make the intermediate steps longer, broader, or trained into the weights.
- Beyond basic CoT: automatic exemplar construction (Auto-CoT), search over reasoning paths (Tree-of-Thought), and sampling-based answer selection (Self-Consistency).

Few-shot vs Zero-shot CoT

- Few-shot CoT provides worked exemplars in the prompt (e.g. "If 6 times 4 equals 24, then what is 6 times 5?" followed by its reasoning steps); zero-shot CoT only appends an instruction such as "Let's think step by step".
- Few-shot buys accuracy at the cost of hand-crafting exemplars for each task; zero-shot avoids that cost but is weaker on hard problems. Auto-CoT aims for few-shot accuracy without the manual effort.



Four prompting settings on the same question: (a) Few-shot, (b) Few-shot-CoT, (c) Zero-shot and (d) Zero-shot-CoT. Only the two CoT settings reach the right answer. Source: Kojima et al. (2022), Fig. 1.

Auto-CoT Prompting

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Automatic Chain of Thought Prompting in Large Language Models

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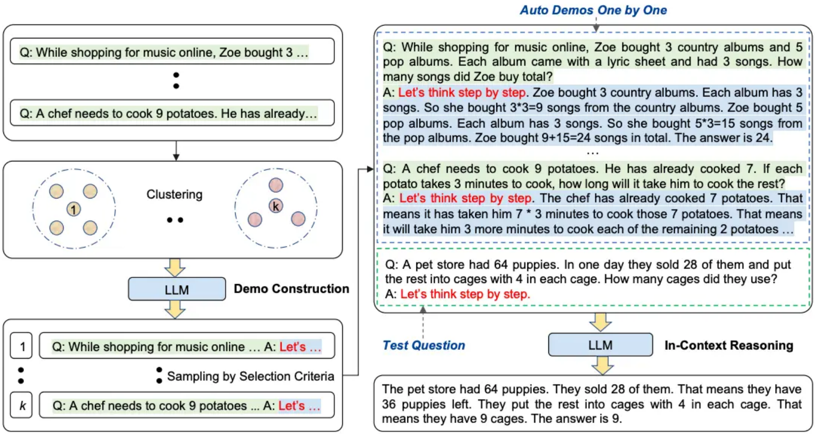
Abstract

Large language models (LLMs) can perform complex reasoning by generating intermediate reasoning steps. Providing these steps for prompting demonstrations is called chain-of-thought (CoT) prompting. CoT prompting has two major paradigms. One leverages a simple prompt like "Let's think step by step" to facilitate step-by-step thinking before answering a question. The other uses a few manual demonstrations one by one, each composed of a question and a reasoning chain that leads to an answer. The superior performance of the second paradigm hinges on the hand-crafting of task-specific demonstrations one by one. We show that such manual efforts may be eliminated by leveraging LLMs with the "Let's think step by step" prompt to generate reasoning chains for demonstrations one by one, i.e., *let's think not just step by step, but also one by one*. However, these generated chains often come with mistakes. To mitigate the effect of such mistakes, we find that diversity matters for automatically constructing demonstrations. We propose an automatic CoT prompting method: Auto-CoT. It samples questions with diversity and generates reasoning chains to construct demonstrations. On ten public benchmark reasoning tasks with GPT-3, Auto-CoT consistently matches or exceeds the performance of the CoT paradigm that requires manual designs of demonstrations. Code is available at `https://github.com/amazon-research/auto-cot`

- **What is Auto-CoT?**
  - A method for building the worked exemplars that few-shot CoT needs, without writing them by hand.
  - The reasoning chains in the prompt are generated by the model itself rather than by an annotator.
- **How it works:**
  - **Cluster:** encode the task's questions with Sentence-BERT and partition them into k clusters.
  - **Construct:** from each cluster take the question closest to the centre and generate its chain with Zero-shot CoT ("Let's think step by step"), keeping the first chain that passes simple length and format criteria.
  - The k resulting question-and-chain pairs become the few-shot prompt. Clustering spreads the exemplars over different question types, so one flawed chain cannot dominate the prompt.

Auto-CoT Prompting (cont.)

- **Benefits:**
  - Removes the manual effort of writing exemplars while matching, and on some benchmarks exceeding, hand-crafted few-shot CoT.
  - Needs only a pool of unlabelled questions from the task: no reference reasoning chains and no answer labels.
- **Cost:**
  - The exemplars are only as good as the zero-shot chains behind them, so the selection criteria are what keep wrong reasoning out of the prompt.



Auto-CoT Prompting. Source: Zhang et al. (2022), Fig. 4.

Auto-CoT Prompting (cont.)

Algorithm 1 Cluster

Require: A set of questions $\mathcal{Q}$ and the number of demonstrations k

Ensure: Sorted questions $\mathbf{q}^{(i)} = [q_1^{(i)}, q_2^{(i)}, \ldots]$ for each cluster i $(i = 1, \ldots, k)$

```
procedure CLUSTER(Q, k)
    for each question q in Q do
        Encode q by Sentence-BERT
    Cluster all the encoded question representations into k clusters
    for each cluster i = 1, ..., k do
        Sort questions q^(i) = [q_1^(i), q_2^(i), ...] in the ascending order of the distance to the cluster center
    return q^(i) (i = 1, ..., k)
```

Clustering step: group the task's questions and sort each cluster by distance to its centre. Source: Zhang et al. (2022), Alg. 1.

Auto-CoT Prompting (cont.)

Algorithm 2 Construct

Require: Sorted questions $\mathbf{q}^{(i)} = [q_1^{(i)}, q_2^{(i)}, \ldots]$ for each cluster i $(i = 1, \ldots, k)$, empty demonstration list $\mathbf{d}$

Ensure: Demonstration list $\mathbf{d} = [d^{(1)}, \ldots, d^{(k)}]$

```
procedure CONSTRUCT(q^(i), ..., q^(k))
    for each cluster i = 1, ..., k do
        for each question q_j^(i) in q^(i) do
            Generate rationale r_j^(i) and answer a_j^(i) for q_j^(i) using Zero-Shot-CoT
            if q_j^(i), r_j^(i) satisfy selection criteria then
                Add d^(i) = [Q: q_j^(i), A: r_j^(i) ∘ a_j^(i)] to d
                break
    return d
```

Construction step: generate a chain per cluster with Zero-shot CoT and keep the first that passes the selection criteria. Source: Zhang et al. (2022), Alg. 2.

Tree of Thoughts: Deliberate Problem Solving with Large Language Models

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Abstract

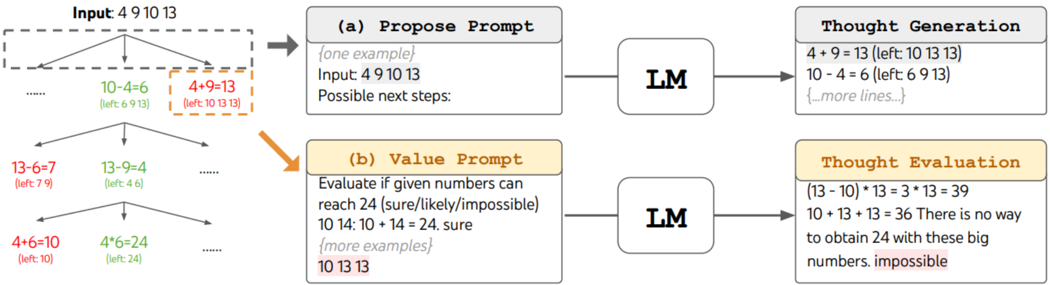
Language models are increasingly being deployed for general problem solving across a wide range of tasks, but are still confined to token-level, left-to-right decision-making processes during inference. This means they can fall short in tasks that require exploration, strategic lookahead, or where initial decisions play a pivotal role. To surmount these challenges, we introduce a new framework for language model inference, "Tree of Thoughts" (ToT), which generalizes over the popular "Chain of Thought" approach to prompting language models, and enables exploration over coherent units of text ("thoughts") that serve as intermediate steps toward problem solving. ToT allows LMs to perform deliberate decision making by considering multiple different reasoning paths and self-evaluating choices to decide the next course of action, as well as looking ahead or backtracking when necessary to make global choices. Our experiments show that ToT significantly enhances language models' problem-solving abilities on three novel tasks requiring non-trivial planning or search: Game of 24, Creative Writing, and Mini Crosswords. For instance, in Game of 24, while GPT-4 with chain-of-thought prompting only solved 4% of tasks, our method achieved a success rate of 74%. Code repo with all prompts: `https://github.com/princeton-nlp/tree-of-thought-llm`.

- **What is ToT?**
  - A reasoning technique that structures the thought process as a tree, allowing for exploration of multiple reasoning paths.
  - Each node in the tree represents a step in the reasoning process, and branches represent different possible paths or decisions.
- **How it works:**
  - The model generates a tree structure where each branch represents a different reasoning path.
  - The model can explore multiple paths simultaneously, evaluating the outcomes of each.

Tree-of-Thought (ToT) Reasoning (cont.)

- **Benefits:**
  - Allows for more complex reasoning by considering multiple possibilities at once.
  - Helps the model avoid getting stuck in a single line of reasoning, improving problem-solving capabilities.
- **Example (Game of 24):**
  - Given four numbers, reach 24 using each exactly once. A thought is one intermediate equation, e.g. "$4 + 9 = 13$ (left: 5, 10, 13)".
  - At each level the model proposes candidate next equations, then rates each resulting state as sure, maybe, or impossible, and only the most promising states are expanded.
  - Search matters here because a single greedy chain commits to its first arithmetic step and cannot recover: CoT solves 4% of these puzzles, ToT with breadth 5 solves 74%.

Tree-of-Thought (ToT) Reasoning (cont.)



ToT in a game of 24. The LM is prompted for (a) thought generation and (b) valuation. Source: Yao et al. (2023), Fig. 2.

Tree-of-Thought (ToT) Reasoning (cont.)

Algorithm 1 ToT-BFS($x, p_\theta, G, k, V, T, b$)

Require: Input x, LM p_θ, thought generator $G()$ & size limit k, states evaluator $V()$, step limit T, breadth limit b.

$S_0 \leftarrow \{x\}$

for $t = 1, \cdots, T$ **do**

  $S'_t \leftarrow \{[s, z] \mid s \in S_{t-1}, z_t \in \mathrm{G}(p_\theta, s, k)\}$

  $V_t \leftarrow V(p_\theta, S'_t)$

  $S_t \leftarrow \arg\max_{S \subset S'_t, |S|=b} \sum_{s \in S} V_t(s)$

end for

return $G(p_\theta, \arg\max_{s \in S_T} V_T(s), 1)$

Breadth-first search over thoughts: keep the b best-valued states at each of the T levels. Source: Yao et al. (2023), Alg. 1.

Tree-of-Thought (ToT) Reasoning (cont.)

Algorithm 2 ToT-DFS($s, t, p_\theta, G, k, V, T, v_{th}$)

Require: Current state s, step t, LM p_θ, thought generator $G()$ and size limit k, states evaluator $V()$, step limit T, threshold v_{th}

```
if t > T then record output G(p_θ, s, 1)
end if
for s' ∈ G(p_θ, s, k) do                          ▷ sorted candidates
    if V(p_θ, {s'})(s) > v_thres then             ▷ pruning
        DFS(s', t + 1)
    end if
end for
```

Depth-first search over thoughts: follow one branch, pruning any state valued below the threshold v_{th}.
Source: Yao et al. (2023), Alg. 2.

Tree-of-Thought (ToT) Reasoning (cont.)

Method	Success
IO prompt	7.3%
CoT prompt	4.0%
CoT-SC (k=100)	9.0%
ToT (ours) (b=1)	45%
ToT (ours) (b=5)	**74%**
IO + Refine (k=10)	27%
IO (best of 100)	33%
CoT (best of 100)	49%

Table 2: Game of 24 Results.

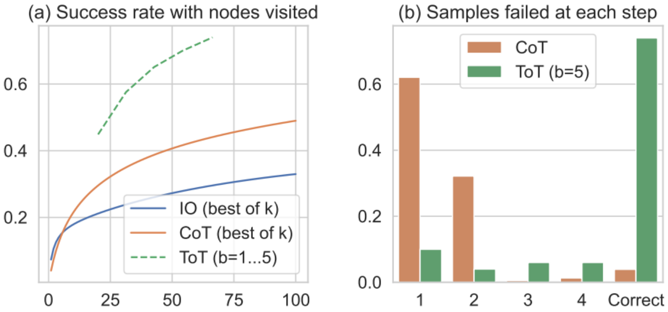


Figure 3: Game of 24 (a) scale analysis & (b) error analysis.

Source: Yao et al. (2023), Table 2 and Fig. 3.

Tree-of-Thought (ToT) Reasoning (cont.)

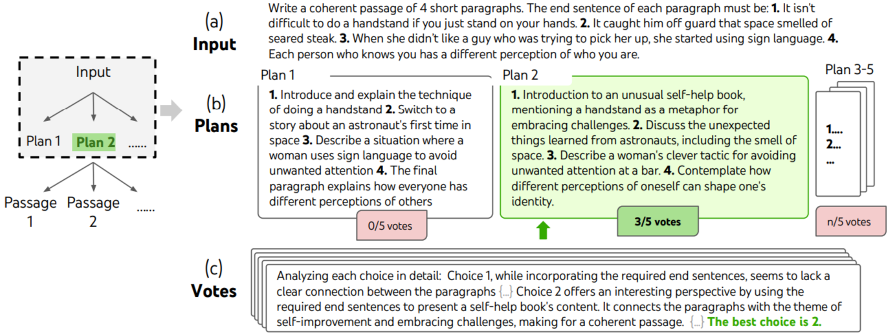


Figure 4: A step of deliberate search in a randomly picked Creative Writing task. Given the input, the LM samples 5 different plans, then votes 5 times to decide which plan is best. The majority choice is used to consequently write the output passage with the same sample-vote procedure.

Source: Yao et al. (2023), Fig. 4.

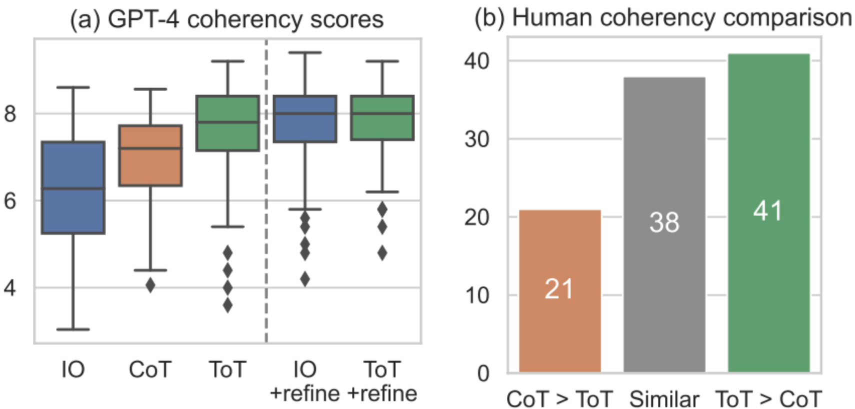


Figure 5: Creative Writing results.

Source: Yao et al. (2023), Fig. 5.

Self-Consistency (SC)

- Self-consistency samples several reasoning paths for the same prompt, then keeps the answer the paths agree on.
- The figures below give the empirical picture from Wang et al. (2022): consistent gains across arithmetic and commonsense benchmarks, growing with the number of sampled paths, at a linear increase in inference cost.
- Sampling many paths at inference time is an early form of test-time compute scaling, the same trade later used by dedicated reasoning models.

Self-Consistency (SC) (cont.)

SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT REASONING IN LANGUAGE MODELS

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ABSTRACT

Chain-of-thought prompting combined with pre-trained large language models has achieved encouraging results on complex reasoning tasks. In this paper, we propose a new decoding strategy, *self-consistency*, to replace the naive greedy decoding used in chain-of-thought prompting. It first samples a diverse set of reasoning paths instead of only taking the greedy one, and then selects the most consistent answer by marginalizing out the sampled reasoning paths. Self-consistency leverages the intuition that a complex reasoning problem typically admits multiple different ways of thinking leading to its unique correct answer. Our extensive empirical evaluation shows that self-consistency boosts the performance of chain-of-thought prompting with a striking margin on a range of popular arithmetic and commonsense reasoning benchmarks, including GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%) and ARC-challenge (+3.9%).

Self-Consistency (SC) (cont.)

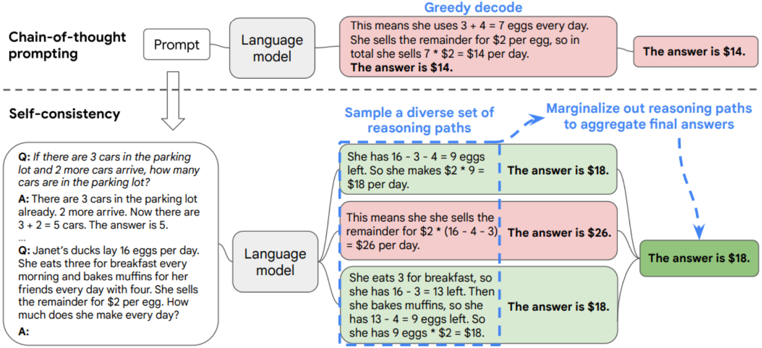


Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the "greedy decode" in CoT prompting by sampling from the language model's decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

Source: Wang et al. (2022), Fig. 1.

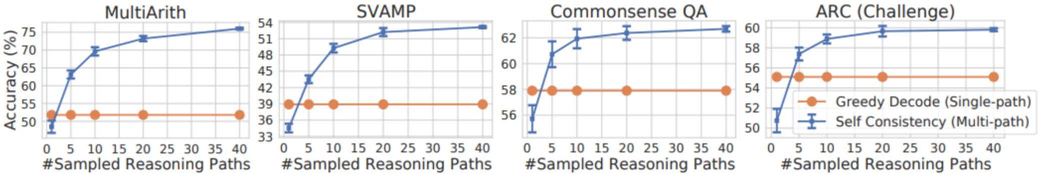


Figure 2: Self-consistency (blue) significantly improves accuracy over CoT-prompting with greedy decoding (orange) across arithmetic and commonsense reasoning tasks, over LaMDA-137B. Sampling a higher number of diverse reasoning paths consistently improves reasoning accuracy.

Source: Wang et al. (2022), Fig. 2. Note that the gain saturates: extra samples buy less and less.

Self-Consistency (SC) (cont.)

	GSM8K	StrategyQA
Question	*Henry made two stops during his 60-mile bike trip. He first stopped after 20 miles. His second stop was 15 miles before the end of the trip. How many miles did he travel between his first and second stops?*	*Yes or no: Is Albany, Georgia the most populous US Albany?*
Greedy Decode	He traveled 60 miles in total. He stopped after 20 miles, so he traveled 60 - 20 = 40 miles between the first and second stops. The answer is 40.	Albany, Georgia is the most populous US Albany. Thus, the answer is yes.
Sampled Path 1	Henry travelled a total of 60 miles. His first stop was 20 miles in, and his second stop was 15 miles before the end. So between his first and second stops he travelled 60 - 20 -15 = 25 miles. The answer is 25.	The most populous US Albany is Albany, New York. Thus, Albany, Georgia is not the most populous US Albany. So the answer is no.
Sampled Path 2	He made two stops during a 60-mile trip. The first was 20 miles into the trip. The second was 15 miles before the end of the trip. This means the second stop was 60 - 15 = 45 miles into the trip. Since he made the stops in order, the second stop must have been 45 - 20 = 25 miles after the first stop. The answer is 25.	Albany, Georgia has a population of about 88,000. Albany, New York has a population of about 95,000. Thus, Albany, Georgia is not the most populous US Albany. So the answer is no.

Table 4: Examples where self-consistency helps repair the errors over greedy decode, on PaLM-540B. Two sampled reasoning paths that are consistent with the ground truth are shown.

Source: Wang et al. (2022), Table 4.

Self-Consistency (SC) (cont.)

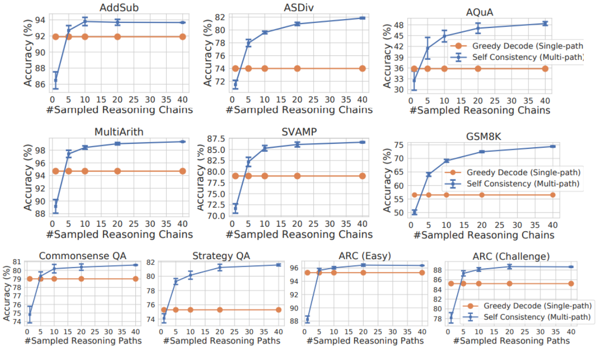
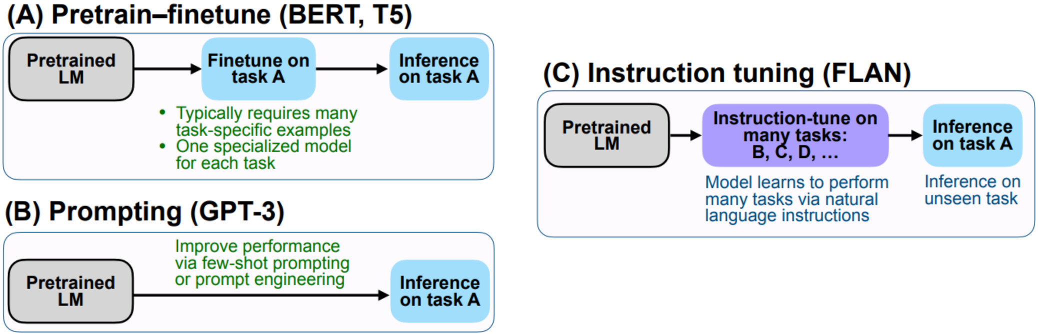


Figure 8: Self-consistency (blue) significantly improves accuracy across various arithmetic and commonsense reasoning tasks, over PaLM-540B. Sampling a higher number of diverse reasoning paths consistently helps reasoning accuracy.

Source: Wang et al. (2022), Fig. 8.

Instruction Tuning for Reasoning Tasks



Instruction tuning next to the two paradigms it draws on: task-specific finetuning and few-shot prompting. Source: Wei et al. (2022), Fig. 2.

Instruction Tuning for Large Language Models: A Survey

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Abstract

This paper surveys research works in the quickly advancing field of instruction tuning (IT), a crucial technique to enhance the capabilities and controllability of large language models (LLMs). Instruction tuning refers to the process of further training LLMs on a dataset consisting of (INSTRUCTION, OUTPUT) pairs in a supervised fashion, which bridges the gap between the next-word prediction objective of LLMs and the users' objective of having LLMs adhere to human instructions. In this work, we make a systematic review of the literature, including the general methodology of IT, the construction of IT datasets, the training of IT models, and applications to different modalities, domains and application, along with analysis on aspects that influence the outcome of IT (e.g., generation of instruction outputs, size of the instruction dataset, etc). We also review the potential pitfalls of IT along with criticism against it, along with efforts pointing out current deficiencies of existing strategies and suggest some avenues for fruitful research. ↥ ↥ ↥ ↥

Instruction Tuning (cont.)

- **What is Instruction Tuning?**
  - Fine-tuning LLMs on a dataset of instruction-response pairs.
  - Enhances the model's ability to follow instructions and perform tasks.
- **Why is it important?**
  - Improves model performance on diverse tasks.
  - Enables better generalization to unseen tasks.

Why Instruction Tune LLMs?

- Pre-trained LLMs are optimized for next-word prediction, not for following instructions or engaging in conversations.
- Without instruction tuning, LLMs may generate text that is linguistically correct but not aligned with user intent.
- Instruction tuning uses supervised learning on instruction-response pairs, teaching the model to generate helpful and relevant outputs for user prompts.
- Fine-tuning is much more efficient than training a large model from scratch and can be performed with less data and compute.
- Instruction tuning bridges the gap between generic language modeling and practical, task-oriented use cases, making LLMs more useful and predictable.

Summary of Instruction Tuning

- **Definition:** Fine-tuning LLMs on diverse instruction-response pairs.
- **Key Elements:**
  - *Instruction:* Task description in natural language.
  - *Additional Information:* Optional context or input.
  - *Desired Output:* Target response for evaluation.
- **Benefits:**
  - Improves ability to follow instructions.
  - Reduces need for prompt engineering and few-shot examples.
  - Enhances generalization to unseen tasks.
- **Applications:**
  - Translation, question answering, reading comprehension, NLI.
  - Broad improvements across reasoning and practical tasks.

Examples of Instruction Tuning in Practice

- **FLAN-T5:**
  - Instruction-tuned T5 models show strong improvements on reasoning benchmarks such as DROP (reading comprehension), ARC (science questions), and MultiRC (multi-sentence reasoning).
  - For example, when given instructions like “Answer the following science question” or “Read the passage and answer the question,” FLAN-T5 produces more accurate and relevant responses.
  - *Example:*
    - **Instruction:** “Read the passage and answer: What did the boy do after school?”
    - **Passage:** “The boy finished his homework and went to play soccer.”
    - **FLAN-T5 Output:** “He went to play soccer.”

Instruction Tuning (cont.)

- **FLAN-MoE:**
  - Mixture-of-Experts (MoE) models, when instruction-tuned, outperform larger dense models on reasoning tasks while using less computation: FLAN-MoE-32B beats FLAN-PaLM-62B on four benchmarks at a third of the FLOPs.
  - Sparsity and instruction tuning compound: the gain from instruction tuning is larger for MoE models than for their dense counterparts.
  - *Example:*
    - **Instruction:** "Translate this sentence to French: The cat is sleeping."
    - **FLAN-MoE Output:** "Le chat dort."

Why Does Instruction Tuning Work?

- Models learn to **follow instructions step by step**, rather than just predicting the next word or memorizing answers.
- Encourages **logical reasoning**: Models are guided to explain their answers or show their work, which helps in tasks like math, logic puzzles, and multi-step questions.
- **Generalization:** Instruction-tuned models can handle new types of questions or tasks they have not seen before, simply by following the provided instruction.
- **Variety of Examples:**
  - *Math:* "Solve: What is 15 divided by 3?" → "5"
  - *Reasoning:* "Is a whale a mammal or a fish? Explain your answer." → "A whale is a mammal because it gives birth to live young and breathes air."
  - *Summarization:* "Summarize the following paragraph in one sentence."

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Instruction Tuning vs. Multi-Task Fine-Tuning (FLAN Ablation Study)

- **Research Question:** Are the benefits of instruction tuning due to the instructions themselves, or just from fine-tuning on multiple tasks?
- **Ablation Setups:**
  - *No Template:* Only input-output pairs, e.g., input: "the dog runs", output: "le chien court".
  - *Dataset Name:* Input prepended with task and dataset, e.g., "[Translation: WMT 14 to French] The dog runs".
  - *FLAN Instructions:* Full natural language instruction, e.g., "Please translate this sentence to French: 'The dog runs.' "
- **Results:** zero-shot accuracy averaged over four held-out task clusters.
  - FLAN instructions 55.2%, dataset name 46.6%, no template 37.3%: a gap of 8.6 and 17.9 points respectively.
  - **Conclusion:** Most of the benefit comes from the instructions themselves, not from multi-task fine-tuning alone.

Chain-of-Thought (CoT) Fine-Tuning

- **How is it done?** This can be achieved by:
  - Few-shot prompting with exemplars that demonstrate sequential reasoning.
  - Appending phrases like "think step by step" to prompts.
- **Why is it important?**
  - CoT prompting significantly enhances zero-shot performance on arithmetic, symbolic, and logical reasoning tasks.
  - Research (Chung et al., 2022) shows that instruction tuning without CoT tasks degrades performance on CoT evaluations.
  - Including CoT tasks in instruction tuning improves performance across all evaluations.
- **Key Insight:** Fine-tuning on CoT tasks, both with and without few-shot exemplars, enables models to better perform step-by-step reasoning, rather than jumping to linguistically plausible answers.

Instruction Tuning a Small Model on a Large One's Reasoning

- FLAN instruction-tunes on human-written task templates. A cheaper route is to let a strong model write the training data: prompt a teacher LLM with Zero-shot CoT, then instruction-tune a small student on the resulting reasoning chains.
- Ranaldi & Freitas call this *Self-refine Instruction-tuning* and add a second phase: the student generates its own answers and is then tuned on preferences over them with Direct Preference Optimization.
- Both phases matter: adding self-refinement significantly outperforms instruction tuning alone, both on the tuned tasks and on held-out ones, bringing the small students closer to the reasoning ability of the much larger teachers.

Instruction Tuning (cont.)

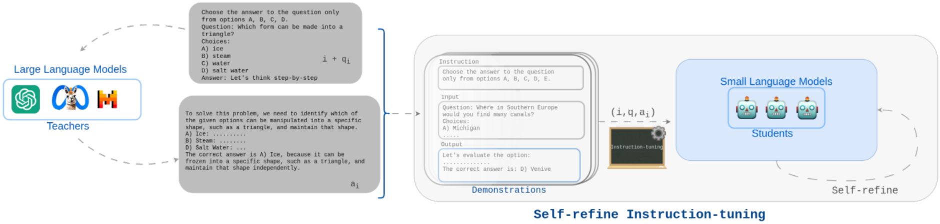

Figure 1: In Self-refine Instruction-tuning, the Demonstrations delivered by teacher models are used to align reasoning abilities in a teacher-student setting. Following the transference of step-wise reasoning knowledge via instruction tuning, the students Self-refine their abilities with the support of Direct Preference Optimization methods.

Source: Ranaldi & Freitas (2024), Fig. 1.

Instruction Tuning (cont.)

Prompt for task: OBQA, CSQA, PIQA, SIQA	*Prompt for task:* GSM8k, MultiArith
Choose the answer to the question only from options A, B, C, [...]. **Question:** <Question> **Choices:** A) <Option1> B) <Option2> C) <Option3> .... **Answer:**	**Answer the following mathematical question with numerical solution.** **Question:** <Question> **Answer:**

Table 2: Example of input-prompt for multiple-choices (left) and mathematical (right) question-answering benchmarks.

Prompt for task: OBQA, CSQA, PIQA, SIQA	*Prompt for task:* GSM8k, MultiArith
Choose the answer to the question only from options A, B, C, [...]. **Question:** <Question> **Choices:** A) <Option1> B) <Option2> C) <Option3> .... **Answer: Let's think step by step**	**Answer the following mathematical question with numerical solution.** **Question:** <Question> **Answer: Let's think step by step**

Table 3: Example **Zero-shot CoT** of input-prompt for multiple-choices (left) and mathematical (right) question-answering benchmarks (approach used in this work).

Source: Ranaldi & Freitas (2024), Tables 2–3.

Instruction Tuning (cont.)

Instruction

```
Choose the answer to the question only from options A, B, C, D.
```

Input

```
Question: Which form can be made into a triangle?
Choices:
A) ice
B) steam
C) water
D) salt water
Answer: Let's think step-by-step
```

Output (that is the Answer produced by teacher model)

```
Answer:
A) ice: Ice is a solid form of water. In solid form, substances can be molded or shaped into
various forms, including triangles. If you had a block of ice, you could carve or mold it
into a triangular shape.
B) steam: Steam is the gaseous form of water. Being a gas, it doesn't have a fixed shape. You
can't shape steam into a triangle without containing it within some form of solid boundary,
and even then, the steam itself isn't triangular - the container is.
C) water: Water is a liquid that takes the shape of its container. You can pour water into
a triangular container, and while the volume inside would be triangular, the water itself
isn't inherently triangular.
D) salt water: This is essentially the same as option C, but with salt dissolved in it. Being
a liquid, it takes the shape of its container.
Given these considerations, the only substance among the options that can be shaped into a
triangle by itself is A) Ice.
```

Table 6: Example of Demonstration produced by the GPT-3.5 answer on an Open Book Question Answering benchmark instance. The structure is composed by: **Instruction**, **Input** and **Output**.

Source: Ranaldi & Freitas (2024), Table 6.

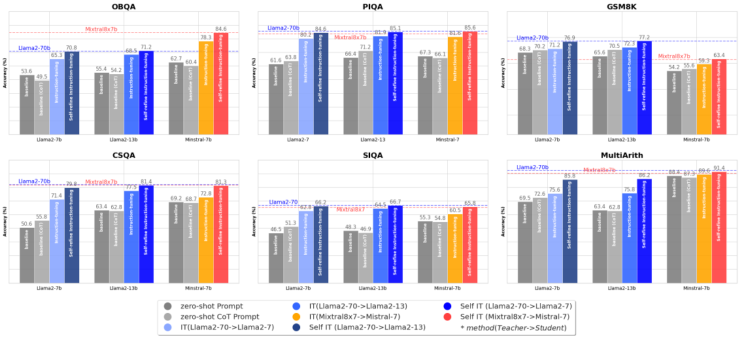

Figure 2: Accuracies (%) on benchmarks (Section 3.1) before Instruction-tuning (i.e., `Baselines` and `Baseline CoT`), after Instruction-tuning (IT) performed on Demonstrations delivering CoT and finally behind the `Self-refine Instruction-tuning` phase (Self IT). In particular, the models were instructed via Demonstrations delivered by in-family LLMs (as described in the legend, we use the notation *method(Teacher->Student)*).

Source: Ranaldi & Freitas (2024), Fig. 2. Dashed lines mark the much larger teacher models.

Challenges and Limitations of Instruction Tuning

- **Dataset Quality and Diversity:**
  - Large, high-quality datasets are costly to create.
  - Reliance on few open-source datasets may reduce diversity.
- **Bias Propagation:**
  - Proprietary LLMs used for instruction generation can spread their biases to open-source models.
- **Evaluation Concerns:**
  - Evaluation often uses proprietary models, which may not reveal all biases or limitations.

Instruction Tuning (cont.)

- **Superficial Learning:**
  - Smaller models may mimic style without true reasoning.
  - Gains may reflect pattern matching, not understanding.
- **Overfitting to Training Tasks:**
  - Evaluation tasks too similar to training can overstate improvements.
- **Key Insight:**
  - Improving base model quality is more impactful than just imitating proprietary systems.

Limitations of Large Reasoning Models

- **Verbose outputs:** Chain-of-Thought and Tree-of-Thought produce long outputs.
  - Extensive reasoning traces can bury the answer the user actually asked for.
- **Prompt sensitivity:** Small changes in wording lead to large changes in output.
  - CoT lifts GSM8K accuracy substantially, but the size of the gain depends on how the exemplars are written.
  - Careful prompt design is therefore part of the method, not an optional extra.
- **Computation overhead:** Worst with self-consistency sampling and Tree-of-Thought search.
  - Cost grows with the number of sampled paths or expanded nodes, so accuracy is bought with inference time.

Can the Reasoning Trace Be Trusted?

The techniques above were each introduced with the claim that visible reasoning makes a model easier to audit. That claim needs qualifying.

- **A trace need not describe the computation.**
  - Turpin et al. (2023) added a biasing feature to the prompt, for example reordering multiple-choice options so the correct answer was always the same letter.
  - Models followed the bias, wrote fluent reasoning that never mentioned it, and lost up to 36 accuracy points across 13 BIG-Bench Hard tasks.
  - So a trace can rationalise an answer after the fact rather than explain how it was produced. It is evidence about what the model will say, not about what it did.

Can the Reasoning Trace Be Trusted? (cont.)

- **Accuracy can collapse rather than degrade.**
  - On puzzles whose difficulty can be dialled up smoothly, Shojaee et al. (2025) report three regimes: plain LLMs win on the easiest instances, reasoning models win in the middle, and both fall to near zero past a threshold.
  - Past that threshold the models spend *fewer* thinking tokens, not more, even with budget left over.
  - Read with care: Lawsen (2025) attributes part of the collapse to output-token limits and to river-crossing instances that are mathematically unsolvable, so the ceiling is not purely a reasoning ceiling.
- **Effort is not well matched to difficulty.**
  - Chen et al. (2024) show o1-style models spending long traces on questions like $2 + 3$, buying no accuracy for the extra tokens.
  - The pathology runs both ways: overthinking on the easy end, and giving up early on the hard end.

Can the Reasoning Trace Be Trusted? (cont.)

- **What follows for practice**
  - Check the answer, not the prose: where correctness matters, verify the output against something external such as a unit test, a solver, or a known result.
  - Do not present a trace to a user as an explanation of the model's decision unless it has been tested for faithfulness.
  - Automatic checking of answers is exactly what the reinforcement-learning approach covered later builds on, and it is the reason that approach is more reliable than prompting alone.

Summary

- **Large reasoning models** differ from pattern-matching LLMs: they spend test-time compute on explicit multi-step reasoning traces.
- **Prompting techniques** (Chain-of-Thought, Tree-of-Thought, Self-Consistency) elicit reasoning from capable models without any training.
- **Instruction tuning** (FLAN and successors) teaches models to follow task descriptions and generalize to unseen tasks; CoT fine-tuning bakes reasoning into the weights.
- The leading systems (o1/o3, DeepSeek-R1, Gemini Flash Thinking, QwQ) combine these ingredients with RL post-training.
- Costs remain: verbose traces, prompt sensitivity, and inference compute overhead.
- **A trace is not a proof.** Models can write reasoning that omits what actually drove the answer, so verify the answer rather than trusting the prose.

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Limits of reasoning traces

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